

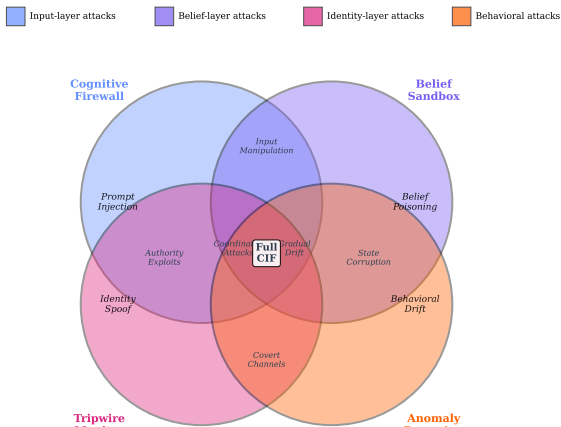
Discussion

Synthesis of Findings

Our empirical evaluation across six production multiagent architectures validates the core theoretical claims of the Cognitive Integrity Framework (Part 1):

Why Layered Defense Succeeds

Defense Mechanism Detection Overlap



Limitations

Detection Gaps Remaining

Despite strong overall performance, specific attack types remain challenging:

- ▶ **Semantic equivalent attacks:** Rephrased injections that preserve meaning evade pattern-matching defenses. Future work should incorporate semantic understanding into the firewall.
- ▶ **Progressive drift:** Sub-threshold belief changes accumulate below detection windows. Longer observation windows trade off against response latency.
- ▶ **Orchestrator compromise:** Outside our threat model assumption (honest orchestrator). Multi-orchestrator architectures provide potential mitigation.

Scalability Constraints

Our evaluation focused on systems with 3-10 agents. Scaling considerations include:

- ▶ Consensus latency grows quadratically with agent count
- ▶ Provenance depth in deep chains slows verification

Relationship to Prior Work

CIF extends prior work in several directions:

- ▶ **Prompt injection defenses:** Existing approaches focus on single-agent scenarios; CIF addresses inter-agent attack propagation
- ▶ **Byzantine fault tolerance:** Classical BFT assumes crash or arbitrary faults; CIF addresses cognitive manipulation specifically
- ▶ **Trust frameworks:** Prior trust systems lack the bounded delegation guarantees that prevent amplification

Future Directions

Adaptive Defenses

Detection rates degrade as adversaries learn to evade (see detection degradation analysis in Part 1, Section 4). Future work should explore:

- ▶ Adversarial retraining of detection mechanisms
- ▶ Honeypot agents to detect novel techniques
- ▶ Formal safety margins for bounded detection degradation

Emergent Behavior Security

As multiagent systems scale, emergent collective behaviors become security-relevant:

- ▶ Formal characterization of “safe” emergent properties
- ▶ Detection of emergent coordination indicating compromise
- ▶ Sandboxing that preserves beneficial emergence

Cross-System Federation

Current CIF deployment assumes a single operator. Future work should address:

- ▶ Federated trust across organizational boundaries